



Spec Driven Development

using CC-SDD

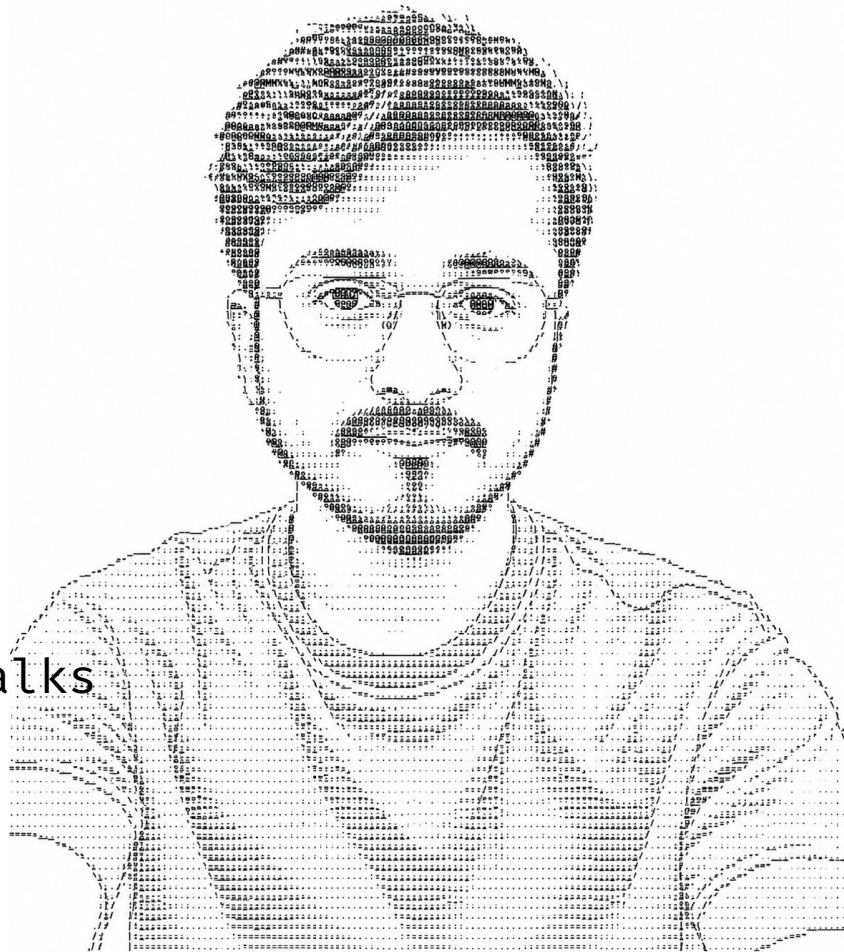
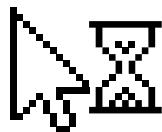


By Taosif Jamal



About Me

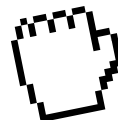
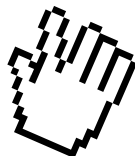
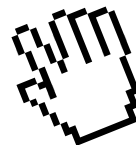
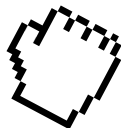
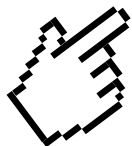
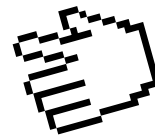
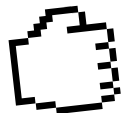
- > A Curious **Engineer**
- > Works at **Scapia**
- > **Indie App Dev** having app with 500K Downloads
- > **Flutter Pub** community manager
- > **Active contributor** of Flutter community via Code, Articles & Talks





01

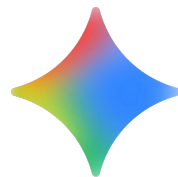
Show of Hands





Show of Hands

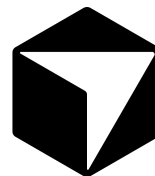
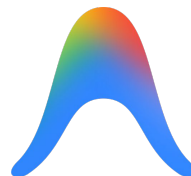
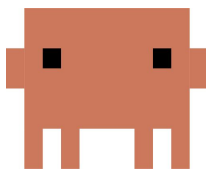
How many of you have **copy pasted code** from these?





Show of Hands

How many of you **use AI IDEs** such as these:





Show of Hands

How many of you have an AI tool **subscription**?





02

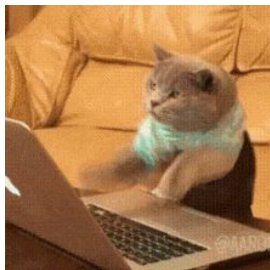
We're Bad Vibe Coders



We've all done this!!!

Rush

We rush through AI
code generation
and miss out on
important
decisions





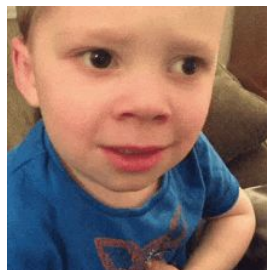
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Rush

We rush through AI
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Alien

We don't fully
understand what AI
is writing. We
Realise it after
code generation





We've all done this!!!

Rush

We rush through AI
code generation
and miss out on
important
decisions

Alien

We don't fully
understand what AI
is writing. We
Realise it after
code generation

Waste

We waste through
lot of time and
tokens trying to
fix and get
desired results





So many problems in life

Unplanned

We don't know what AI will
write until it's done



Code Drift

Written solution is
different from requirements

Wrong Style

Written code is not in
project style

Context

No Context of internal or
domain knowledge

No Continue

Difficult to continue from
middle if it's a large task

Token Burn

We end up wasting lots of
tokens trying to fix
something we didn't want



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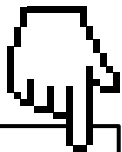
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What do we want?







How do we get it?



A ○





03

The System



Planning

What should be built

Defining

How it should be built

Validating

What is getting built



CC-SDD

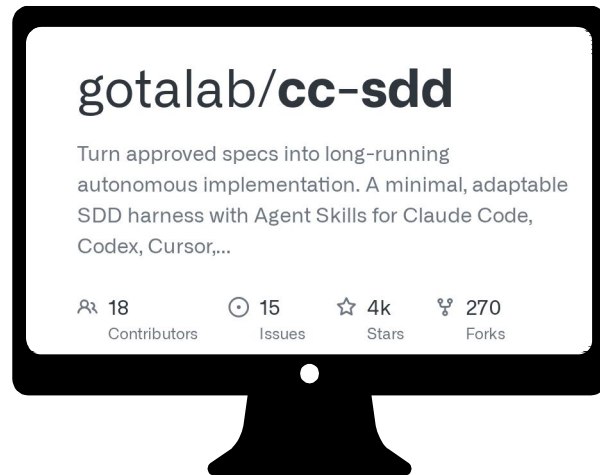
Claude Code – Spec Driven Development
(Inspired from Kiro by Amazon)



What is CC-SDD

OpenSource **Development Workflow**
For AI Agents

It ensures that AI agents write and implement code based on strictly **human-approved specifications** (such as requirements, design, and tasks) rather than just guessing based on broad prompts





What are the specs

- ▼ .kiro
- ▼ settings / templates
- ▼ specs
 - ↓ bugfix-init.md U
 - ↓ bugfix.md U
 - ↓ decisions.md U
 - ↓ design-hld.md U
 - ↓ design-lll.md U
 - ↓ design-qa-log.md U
 - ↓ design.md U
 - { } init.json U
 - ↓ learning-recorder-prompt.md U
 - ↓ learnings.md U
 - ↓ patterns.md U
 - ↓ requirements-init.md U
 - ↓ requirements.md U
 - ↓ research.md U
 - ↓ tasks.md U





How are we building?

01 // DISCOVERY

/kiro-discovery

Understands scope of task
in hand



02 // INIT

/kiro-spec-init

Creates a spec file so it
tracks phases



03 // REQUIREMENTS

/kiro-spec-reqs

Understand requirement



04 // DESIGN

/kiro-spec-design

Design HLD+LLD



05 // TASKS

/kiro-spec-tasks

Generate checklist



06 // IMPLEMENT

/kiro-impl

Implement the tasks



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Design HLD+LLD





How are we building?

01 // DISCOVER

/kiro-discover

Understands scope
in hand

/kiro-discovery

Explores and scopes the
task to understand what
needs to be built.

REQUIREMENTS

spec-reqs

and requirement



DESIGN

spec-design

LD+LLD

06 // IMPLEMENT

/kiro-impl

Implement the

OK





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Implement the tasks



How are we building?

01 // DISCOVER

/kiro-discover

Understands schema
in hand

/kiro-spec-init

Initializes a spec file to
track progress through the
phases.

REQUIREMENTS

spec-reqs

and requirement



DESIGN

spec-design

LD+LLD

06 // IMPLEMENT

/kiro-impl

Implement the

NEXT





How are we building?

01 // DISCOVER

/kiro-discover

Understands sc
in hand

Spec

= What +
Requirements
Boundaries

Plan

= the “How”;
implementation
approach

REQUIREMENTS

spec-reqs

nd requirement



DESIGN

spec-design

LD+LLD

06 // IMPL

/kiro-impl

Implement the

OK





How are we building?

01 // DISCOVERY

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Understands scope of task
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Creates a spec file so it
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Implement the tasks



05 // TASKS

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Design HLD+LLD



How are we building?

01 // DISCOVER

/kiro-discover

Understands sc
in hand

/kiro-spec-requirements

Captures and clarifies the
functional/non-functional
requirements.

REQUIREMENTS

spec-reqs

nd requirement



06 // IMPLEMENT

/kiro-impl

Implement the

DESIGN

spec-design

LD+LLD

OK



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Design HLD+LLD



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Generate checklist



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Implement the tasks





How are we building?

01 // DISCOVER

/kiro-discover

Understands sc
in hand

/kiro-spec-design

Produces the high-level and
low-level design (HLD +
LLD) for the solution.

REQUIREMENTS

spec-reqs

nd requirement



DESIGN

spec-design

LD+LLD

06 // IMPLME

/kiro-impl

Implement the

OK





How are we building?

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Understands scope of task
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02 // INIT

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Creates a spec file so it
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Understand requirement



04 // DESIGN

/kiro-spec-design

Design HLD+LLD



05 // TASKS

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Generate checklist



06 // IMPLEMENT

/kiro-impl

Implement the tasks





How are we building?

01 // DISCOVER

/kiro-discover

Understands sc
in hand

/kiro-spec-tasks

Breaks the design down
into a concrete, checkable
task list.

REQUIREMENTS

spec-reqs

nd requirement



DESIGN

spec-design

LD+LLD

06 // IMPLEMENT

/kiro-impl

Implement the

OK





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06 // IMPLEMENT

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Implement the tasks





How are we building?

01 // DISCOVER

/kiro-discover

Understands sc
in hand

/kiro-impl

Executes implementation
against the generated task
checklist.

06 // IMPL

/kiro-impl

Implement the

REQUIREMENTS

spec-reqs

nd requirement



DESIGN

spec-design

LD+LLD

OK





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Design HLD+LLD



05 // TASKS

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Generate checklist



06 // IMPLEMENT

/kiro-impl

Implement the tasks



01 Standardised Way of coding

No more wrong style – every output follows your project's conventions

02 No more code drift

Written solution always matches the requirements it came from

03 Fully Planned Before Code

Know exactly what AI will build before it writes a single line

04 Context-Aware Generation

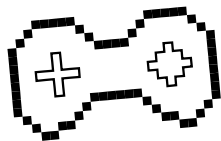
Grounded in your internal and domain knowledge, not guesswork

05 Resumable at Any Point

Pick up large tasks mid-way instead of starting over



Thanks!



Do you have any questions?

@Taosif7 everywhere

